

Dr.Romano part :

1. explain the muscle contraction on microscale (the a,z,h band) and how contraction occurs 🍑.
2. Explain swimming forms(anguliform etc) 🍑
3. Explain Froude number and Strouhal and also Reynold number and write their formulas, what are their units ?
4. What causes thrust in fish (how lift is generated critical angle difference in pressure on airfoil, which area of wing has higher pressure and velocity) 🍑
5. Explain triplasty 🍑
6. Explain acoecolemate, colemate , pseudocolumate what tissues are present and membranes 🍑
7. Explain stigmegey (swarm intelligence part)
8. locomotion vs movement 🍑
9. inchworm locomotion 🍑
10. plant tropism 🍑
11. fatty acids, glucose, fructose
12. serial and parallel robotic arms
13. Automation
14. Explain scale effect (in particular physically, so the mass in a 3d world increase with the cube, while force with square depend on cross sectional area)-
15. Talk about flight (direct vs indirect, neural transmission, draw the muscles of direct vs indirect, other times he asked about the venation of the wings)
16. Talk about jumping(in particular how the locust jump)
17. Related to swimming, how it is possible to move for fishes (the aoa change based on the different bending point)
18. Locust jumping mechanism in detail (all the phases)
19. Stepper motor
20. Scara(in general he asked about all the type of robot and to draw them, RRT TTT RTT ecc)
21. Molecule (draw glucose + fructose and all the functional group, he asked in general about all the functional group)
22. All the hydrodynamic forces that are generated on the fish during swim (buoyancy, thrust, drag and weight)
23. Talk about Fuel cell, battery, flywheels and capacitor.